Auditing a pretrained cube-reorientation policy in simulation

ROB7103/8103 teaching example · Retained Class 06 evidence

A fixed policy reached the first target in 88/100 retained native-MuJoCo trials; 12 timed out.

Question and contribution

How well does the supplied policy reach its first orientation goal under a declared simulation protocol?
This poster analyses retained evaluations; it does not introduce or train the policy.

System and interface

57-value actor observation

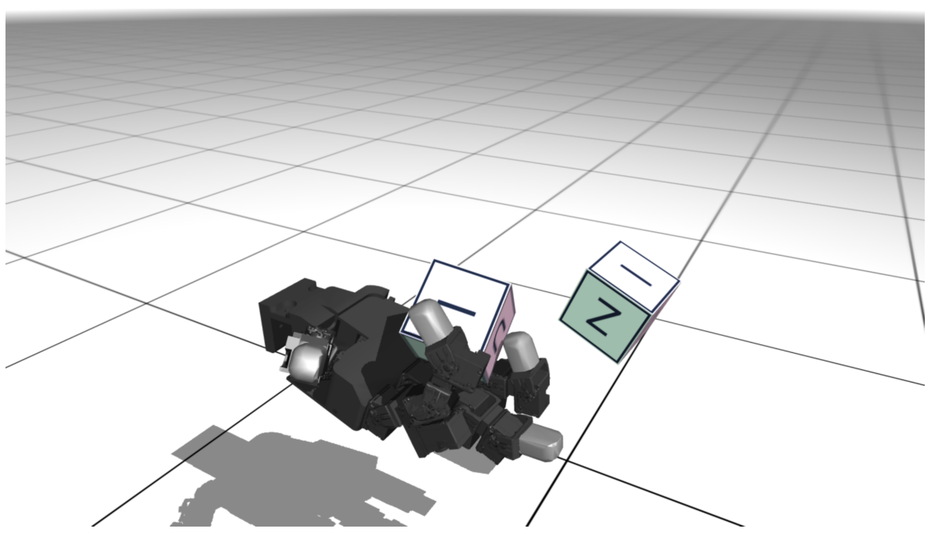
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Supplied pretrained policy

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16 targets → native MuJoCo hand

Task and selected view



Selected simulation frame, seed 9204.
The image illustrates behavior; it is not the 100-trial performance measure.

First-goal evaluation



88 successes · 12 timeouts · 0 falls

100 trials, seeds 20000–20099.
First target below 0.1 rad before fall or 50 s limit. This is the observed result under the stated protocol.

Conditions that define the number

Native-task adapter · MuJoCo 3.6.0 · original observation noise
Physics step 0.01 s · control interval 0.05 s · fixed pretrained actor
All 100 outcomes retained. No training, threshold relaxation or hardware run.

Conclusion and next justified test

The policy is a working, imperfect simulation example under these conditions.
This result does not establish physical-hand reliability. Test one declared change, such as observation delay, using a fresh evaluation seed set.

Sources: Class 06 retained evaluation and selected native rendering.
Policy/task: MuJoCo Playground, commit ef4fefc… · Policy SHA-256: 1893e4a…
Full protocol, source attribution and records accompany the course example.